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Effects of Cottonwood Trees *(Populus angustifolia)* on Yampa River Temperature

Introduction

Rising water temperature is one of the most dangerous threats to the water quality in the Yampa River. This is a direct result of climate change. According to The Nature Conservancy, the Colorado River basin has been in a state of mega-drought for the past twenty years (The Nature Conservancy, 2022). With rising temperatures and less precipitation, especially in the form of snowfall, there is less water in the river basin, resulting in low flows. Less water in the rivers means more evaporation, which leads to even less water. The lower the amount of water in the basin, the more concentrated the hot water becomes, and the hotter the water is the more it evaporates, thus stimulating a self-perpetuating cycle. One of the biggest concerns of increased temperatures in the river is the impact on aquatic life. Hotter water holds less dissolved oxygen, leaving little available for the fish to breathe. The high temperatures also stress the fish, especially during spawning season. Fish species native to the Yampa; the Mountain White Fish, Colorado Pike Minnow, Razorback Sucker, Humpback Chub, and Bonytail, are suffering from increased water temperatures. However, invasive fish species such as the Small Mouth Bass and Northern Pike are tolerant to warmer temperatures, giving them an advantage over their prey, the native fish. (COHeadwaters, 2022).

According to The City of Steamboat Springs Yampa River Management Plan, the river is closed to any recreational activity if the temperature exceeds 75º F (25º C) for two consecutive days, or when flows are below 85 cubic feet per second which shows the correlation of temperature to steam health and maintenance of the Yampa (FOTY, 2024). This is to protect the fish population, as the water is too low for us to responsibly recreate in their environment.

Additionally, hot water is a breeding ground for harmful algae blooms. Assessment of Streamflow and Water Quality in the Upper Yampa River Basin outlined cyanobacteria in Stagecoach Reservoir have been increasingly more abundant in recent years, as a result of land use, hydrology, and climate change. Bloom-forming algae can produce toxins that are harmful to humans and animals, degrade water quality, and decrease water supply. (Day, 2021).

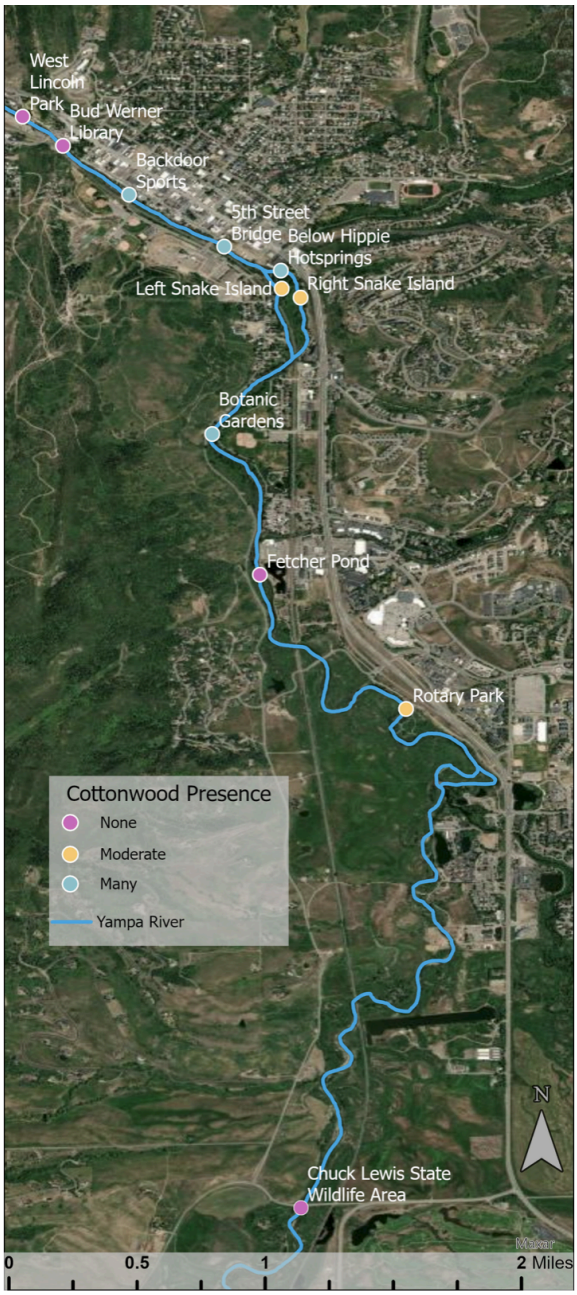
The Yampa River is a perfect candidate for planting *Populus angustifolia* because it is still relatively free-flowing. Research from Anderson's study comparing Cottonwood productivity on the two contrasting dryland river floodplains at Deer Lodge Park along the Yampa and at Brown's Park on the Green River. The study shows that the Basal-Area Increment for Cottonwood trees was higher at Deer Lodge Park than at Brown's Park. The results showed that the Flaming Gorge Dam on the Green River significantly impacted the growth of Cottonwood trees (Andersen, 2016).

Cottonwoods *(Populus angustifolia)* have various benefits to riparian environments. The first is oxygen, the additional trees help create additional oxygen in the surrounding water (Graaff, 2024). Cottonwood trees also help the biodiversity of the areas they are in. They provide habitat for birds and other wildlife and shade that protects native fish from predation. Cottonwoods also produce leaf litter that Macroinvertebrates can feed off. These trees help to mitigate flooding by absorbing excess water. Roots of Cottonwood trees reinforce banks which reduces erosion and sedimentation; this also helps improve overall water quality. *Populus angustifolia* also aids in climate regulation in riparian areas by providing shade and decreasing temperatures. The Yampa River Forest Restoration Project revealed that carbon sequestration among Cottonwood trees is highly efficient due to their large biomass and quick growing period. Carbon sequestration happens during photosynthesis, where the plant takes in carbon dioxide and stores it throughout its leaves, trunk, and roots. This takes carbon dioxide out of the atmosphere, helping to reduce overall climate change (YVSC, 2021).

The primary objective of this study is to find a correlation between the shade provided by Cottonwood trees and reducing the temperature of the water. Hopefully, the results will confirm the efforts made by the Yampa Valley Sustainability Council to show the effectiveness of planting Cottonwood trees along the Yampa River corridor.

Procedure

This experiment was conducted at 11 different locations along the Yampa River near Steamboat Springs, CO (Figure 1). The tests measured temperature (degrees Celsius) and dissolved solids in parts per million (ppm), using a Hofun TDS Meter Digital Water Quality Tester. Special attention was made to ensure the data was collected during similar weather conditions and time of day. At each location, the temperature was taken in water with similar flows and depths to keep data consistent. The locations were based on the presence of cottonwood trees (none, moderate, and many) that were mature enough to cast shade along the river corridor. This was the independent variable that would potentially affect the dependent variable of temperatures and dissolved solids in the water.

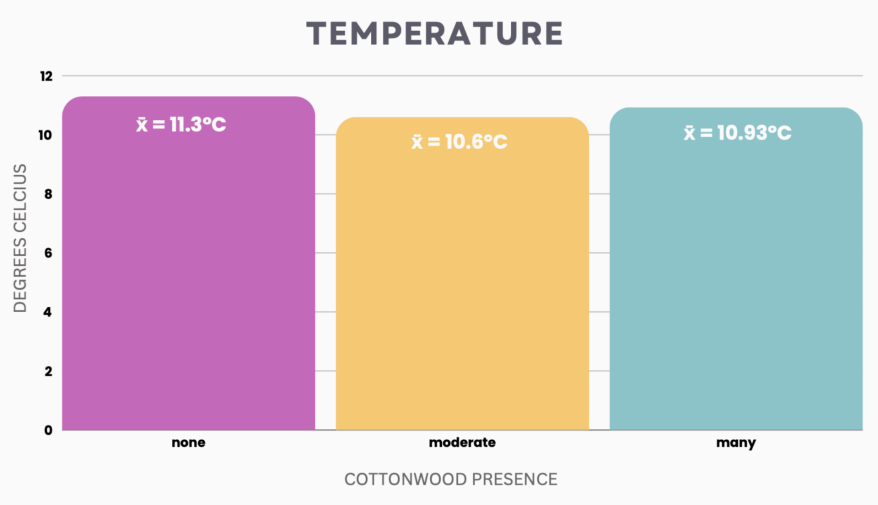


(Figure 1.) Testing sites along the Yampa River in Steamboat Springs, CO.

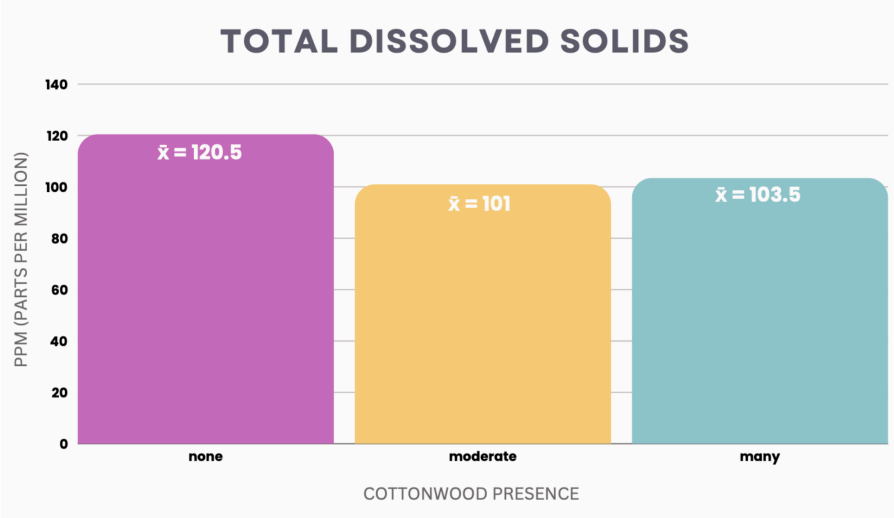
Results

According to the results of my experiment, there is a minor relationship between the presence of *Populus angustifolia* along the river corridor, temperature, and dissolved solids. As for temperature, areas with no cottonwood trees have an average temperature of 11.3ºC (SD 0.69). Areas with moderate cottonwood presence have an average temperature of 10.6ºC (SD 0.79). And areas with many cottonwood trees have a mean temperature of 10.93ºC (SD 0.74). As for dissolved solids (graph 2), areas with no cottonwood trees have a mean value of dissolved solids at 120.5 ppm (SD 24.18). Areas with moderate cottonwood presence have a mean value of 101 ppm (SD 4.35). And areas with many cottonwood trees have a mean value of 103.5 ppm (SD 4.72). After conducting a two-sample t-test with unequal variance on the data for areas with many cottonwoods and no cottonwoods, the results show a p-value of 0.49 for temperature and 0.25 for total dissolved solids.

Graph 1



Graph 2



Discussion

The results of this experiment did not confirm the hypothesis, as the t-test calculations provided values (0.49 and 0.25) which are more than 0.05. This suggests that the data is not statistically significant or that there is a 49% chance the results from the temperature tests are random and a 25% chance from the total dissolved solid test is random. Perhaps the temperature difference among the different locations wasn't substantial because the water is always moving through the river. However, this is even more of a reason to revegetate the entire river corridor (or as much as possible) with cottonwood trees. Additionally, it is possible that other factors could be influencing the data, such as mineral springs runoff in certain testing locations (Hippie Hot Springs, Bud Warner Library, and West Lincoln Park).

If this data were to be collected again, dissolved oxygen should be assessed instead of total dissolved solids. Although evaluating dissolved solids did provide some indication about the health of the river, dissolved oxygen has a closer relationship to the fish living in the streams and is more closely related to the products of photosynthesis, which *Populus angustifolia* carries out. It would also be more effective to conduct this experiment during the summer when the temperature of the water is more of a concern.

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Although the results did not represent a substantial difference in temperature or total dissolved solids, there is still evidence that cottonwood trees are important for the health of the Yampa River ecosystem. They cast shade upon the water which reduces the water temperature and creates a safe environment for native fish. The efforts made by the Yampa Valley Sustainability Council are vital to restoring the riparian habitat by planting cottonwood trees along the river corridor. Another local organization, Friends of the Yampa, also conducts a river health assessment in different sections of the river every year. These two missions, in addition to citizen science like this project, can help people understand the condition of the river and how to be environmental stewards of land and water.

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